Young 2018 kidney cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

The malignant compartment is a composite of THREE biologically distinct kidney cancers — clear-cell RCC (1,373), Wilms tumour (287), papillary RCC (144). Each sample is (near-)pure for one type (Cramér V(sample,type)=0.93), so integrating “samples” integrates three cancers. Across a θ 0→20 sweep, per-neighbourhood iLISI(tumour-type) stays ~1.0–1.2 vs a 1.64 well-mixed ceiling: the types never intermix at any integration strength. The kNN graph is one component throughout (the misleading scalar), but the embedding shows type-segregated islands.

Gate 2 — module geography → N/A

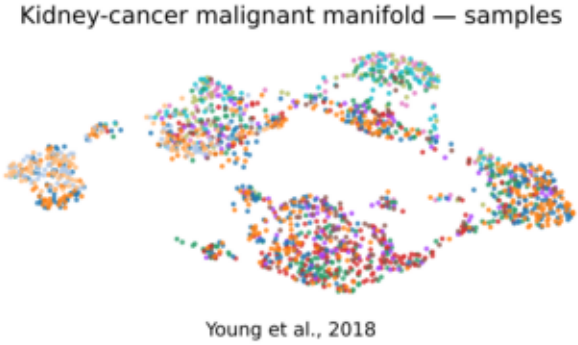
Not evaluated — Gate 1 NO-GO halts the chain before the manifold / signature steps.

Gate 3 — cross-repression → N/A

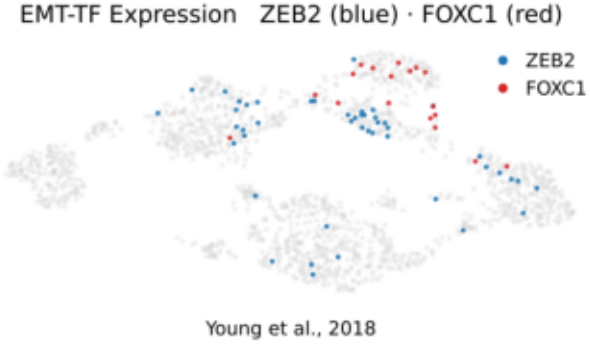
Not evaluated — branching-only, and no trajectory is fit.

STEP FIGURES

Malignant manifold — samples



EMT-TF expression



Dataset.
Kidney cancer (Young et al., 2018), 10x scRNA-seq. 2,055 malignant cells loaded; 1,804 retained after sample thresholding (≥63 cells). A mixture of clear-cell RCC, pediatric Wilms tumour, and papillary RCC.

Verdict.
The embryonic-development cell-state pattern does NOT reproduce: the malignant “manifold” is not one continuous cell-state continuum but three unrelated cancers that refuse to integrate. Gate 1 = NO-GO; no trajectory is fit. Geometry: none (un-integrable composite of three malignancies).

Parameter choices

Sample threshold	≥63 cells (geo-mean of 87→47 gap, 1.85×); 11/25 kept → 1,804 cells	**Overdispersed genes**	531 (scFates find_overdispersed)
Integration	Harmony on sample; θ swept 0–20, read per-neighbourhood; θ=2 for visual	**n_pcs / spread**	20 / 1.3 (10x precedents at ~1.8k cells)
Root / geometry	None — no trajectory fit (Gate 1 NO-GO)	**EMT-TFs shown**	ZEB2 (25.4%) · FOXC1 (9.5%); FOXC/PRRX added, VIM excluded